

Closed-chain kinematics

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Abstract:

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1 Introduction

1.1 Closed-chain mechanisms

Any kinematic structure in which links form one or more closed loops is classified as a closed-chain mechanism. While many traditional robotic limbs employ open-chain configurations, humanoid robots increasingly incorporate closed-chain arrangements in their leg structures to enhance stability, load distribution, and motion precision [1]. These closed-chain systems often resemble parallel mechanisms—structures. In this context, each leg, after the shin, has a kinematic structure as described in the figure 1. The overall configuration forms an inverted modified Stewart-Gough platform. Here, the special emphasis is placed on understanding the geometric and kinematic constraints that arise due to the loop closure conditions inherent in parallel mechanisms.

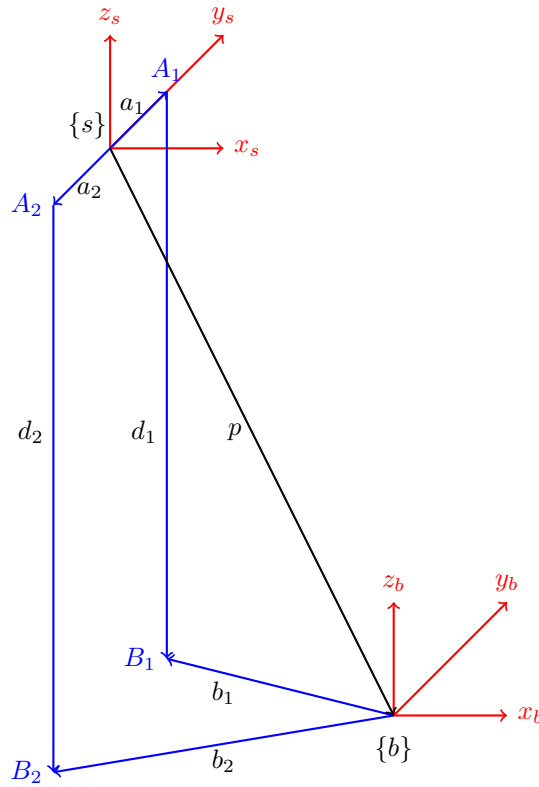


Figure 1: Shin-Foot Mechanism

A fixed frame $\{s\}$ is attached to the knee joint, and body frame $\{b\}$ is assigned to the passive 2D joint at the foot that is in the front. The active prismatic joints are denoted via d_1 and d_2 , and they are connected from A_i to B_i respectively. Frame $\{b\}$, B_1 and B_2 are passive 2D joints on the foot and, A_1 and A_2 are 2D passive joints below the knee.

2 Problem Statement

Given a closed-chain leg structure of a humanoid robot modeled as a parallel mechanism, let $l = (l_1, l_2)$ represent the set of actuated joint variables (in our case, prismatic joints) within each leg chain. We need to solve two problems here:

1. The **forward kinematics problem** consists of determining the position and orientation of the body frame—expressed as a transformation matrix T_{sb} —based on known values of l .
2. Conversely, the **inverse kinematics problem** seeks to compute the required joint variables l that would produce a desired pose of the body frame, given a specific transformation T_{sb} .
3. Jacobian that helps relate the velocity mapping between the two

This formulation must account for the closed-loop constraints imposed by the parallel kinematic structure of the leg mechanism.

3 Solution Approach

One general observation that can be made for serial mechanisms versus parallel mechanisms is the following: for serial chains, the forward kinematics is generally straightforward while inverse kinematics may be complex (e.g., there may be multiple solutions or no solution). For parallel mechanisms, the inverse kinematics is often relatively straightforward (e.g., given the configuration of a platform, it may not be hard to determine the joint variables), while the forward kinematics may be quite complex: an arbitrarily chosen set of joint values may be infeasible or it may correspond to multiple possible configurations of the platform. In this regard, we try to solve the inverse kinematics first.

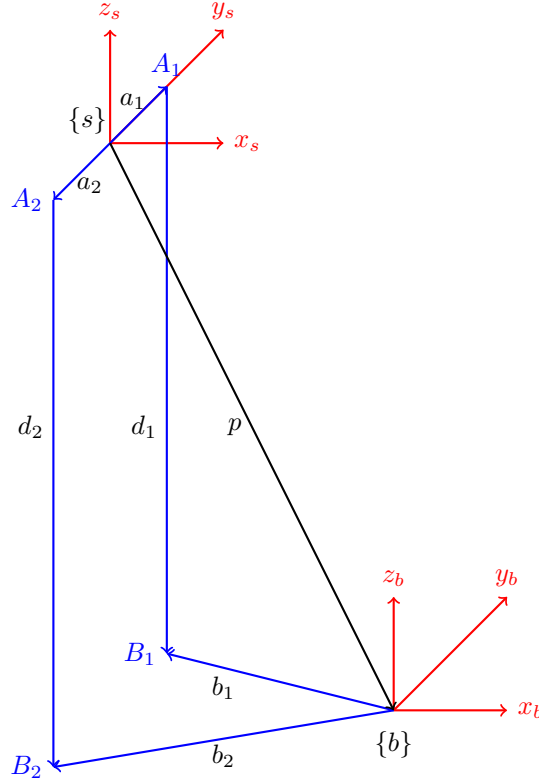


Figure 2: Shin-Foot Mechanism

3.1 Inverse Kinematics

Let p be the vector from the origin of the $\{s\}$ frame to the origin of the $\{b\}$ frame. Let T_{sb} be the transformation matrix from frame $\{s\}$ to $\{b\}$. Further, define the vectors a_i , b_i , and d_i , for $i = 1, 2, 3$, as shown in the figure. From these definitions, it follows clearly that

$$d_i = p + b_i - a_i.$$

Let

$$p = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}, \quad a_i = \begin{bmatrix} a_{ix} \\ a_{iy} \\ a_{iz} \end{bmatrix}, \quad d_i = \begin{bmatrix} d_{ix} \\ d_{iy} \\ d_{iz} \end{bmatrix}, \quad b_i = \begin{bmatrix} b_{ix} \\ b_{iy} \\ b_{iz} \end{bmatrix}$$

where:

- p is the position vector from the origin of the $\{s\}$ frame to the origin of the $\{b\}$ frame, expressed in $\{s\}$ -frame coordinates.
- a_i is the position vector of point a_i expressed in $\{s\}$ -frame coordinates.
- d_i is the vector from a_i to b_i , expressed in $\{s\}$ -frame coordinates.
- b_i is the position vector of point b_i , expressed in $\{b\}$ -frame coordinates.

Note that the components (a_{ix}, a_{iy}, a_{iz}) and (b_{ix}, b_{iy}, b_{iz}) for $i = 1, 2$ are all constants. All vectors are expressed in the $\{s\}$ -frame coordinates, with the exception of b_i , which is originally defined in the $\{b\}$ frame.

Also, given that the translation is constant, and additionally there is no rotation around Z-axis, the transformation matrix $T_s b = R_y R_x$ in regards to following the rpy convention (i.e., rotating around x first and then around y). Thus, to express the final inverse kinematics in the frame $\{s\}$, $\{b_i\}$ must be expressed in $\{s\}$ and thus it follows

$$d_i = p + R_y(\beta)R_x(\alpha)b_i - a_i.$$

where,

$$R_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta_x & -\sin \theta_x \\ 0 & \sin \theta_x & \cos \theta_x \end{bmatrix}, \quad R_y(\beta) = \begin{bmatrix} \cos \theta_y & 0 & \sin \theta_y \\ 0 & 1 & 0 \\ -\sin \theta_y & 0 & \cos \theta_y \end{bmatrix}.$$

Thus the simplified equations for d_1 and d_2 are as follows:

$$d_1 = \begin{bmatrix} -a_{1x} + b_{1x} \cos \beta + b_{1y} \sin \alpha \sin \beta + b_{1z} \cos \alpha \sin \beta + p_x \\ -a_{1y} + b_{1y} \cos \alpha - b_{1z} \sin \alpha + p_y \\ -a_{1z} - b_{1x} \sin \beta + b_{1y} \sin \alpha \cos \beta + b_{1z} \cos \alpha \cos \beta + p_z \end{bmatrix}$$

$$d_2 = \begin{bmatrix} -a_{2x} + b_{2x} \cos \beta + b_{2y} \sin \alpha \sin \beta + b_{2z} \cos \alpha \sin \beta + p_x \\ -a_{2y} + b_{2y} \cos \alpha - b_{2z} \sin \alpha + p_y \\ -a_{2z} - b_{2x} \sin \beta + b_{2y} \sin \alpha \cos \beta + b_{2z} \cos \alpha \cos \beta + p_z \end{bmatrix}$$

And finally, the lengths of the prismatic links are given as follows:

$$l_1 = \|d_1\| = \left\{ (a_{1y} - b_{1y} \cos \alpha + b_{1z} \sin \alpha - p_y)^2 + (-a_{1x} + b_{1x} \cos \beta + b_{1y} \sin \alpha \sin \beta + b_{1z} \cos \alpha \sin \beta + p_x)^2 + (-a_{1z} - b_{1x} \sin \beta + b_{1y} \sin \alpha \cos \beta + b_{1z} \cos \alpha \cos \beta + p_z)^2 \right\}^{1/2}$$

$$l_2 = \|d_2\| = \left\{ (a_{2y} - b_{2y} \cos \alpha + b_{2z} \sin \alpha - p_y)^2 + (-a_{2x} + b_{2x} \cos \beta + b_{2y} \sin \alpha \sin \beta + b_{2z} \cos \alpha \sin \beta + p_x)^2 + (-a_{2z} - b_{2x} \sin \beta + b_{2y} \sin \alpha \cos \beta + b_{2z} \cos \alpha \cos \beta + p_z)^2 \right\}^{1/2}$$

3.2 Jacobian

From the above relation, the length of prismatic links needed to control the orientation of the foot, we can get the jacobian that maps the velocity spaces of the both as follows. The Jacobian is going to be of the form

$$\begin{bmatrix} \dot{l}_1 \\ \dot{l}_2 \end{bmatrix} = \begin{bmatrix} \frac{\partial l_1}{\partial \alpha} & \frac{\partial l_1}{\partial \beta} \\ \frac{\partial l_2}{\partial \alpha} & \frac{\partial l_2}{\partial \beta} \end{bmatrix} \begin{bmatrix} \dot{\alpha} \\ \dot{\beta} \end{bmatrix}$$

Using the chain rule for vector norms, we compute:

$$\frac{\partial l_i}{\partial \theta} = \frac{1}{l_i} \left(d_{ix} \frac{\partial d_{ix}}{\partial \theta} + d_{iy} \frac{\partial d_{iy}}{\partial \theta} + d_{iz} \frac{\partial d_{iz}}{\partial \theta} \right), \quad \theta \in \{\alpha, \beta\}$$

Let us define components of d_1 as:

$$\begin{aligned} d_{1x} &= -a_{1x} + b_{1x} \cos \beta + b_{1y} \sin \alpha \sin \beta + b_{1z} \cos \alpha \sin \beta + p_x \\ d_{1y} &= a_{1y} - b_{1y} \cos \alpha + b_{1z} \sin \alpha - p_y \\ d_{1z} &= -a_{1z} - b_{1x} \sin \beta + b_{1y} \sin \alpha \cos \beta + b_{1z} \cos \alpha \cos \beta + p_z \end{aligned}$$

Then the partial derivatives are:

$$\begin{aligned} \frac{\partial d_{1x}}{\partial \alpha} &= b_{1y} \cos \alpha \sin \beta - b_{1z} \sin \alpha \sin \beta \\ \frac{\partial d_{1y}}{\partial \alpha} &= b_{1y} \sin \alpha + b_{1z} \cos \alpha \\ \frac{\partial d_{1z}}{\partial \alpha} &= b_{1y} \cos \alpha \cos \beta - b_{1z} \sin \alpha \cos \beta \\ \frac{\partial d_{1x}}{\partial \beta} &= -b_{1x} \sin \beta + b_{1y} \sin \alpha \cos \beta + b_{1z} \cos \alpha \cos \beta \\ \frac{\partial d_{1y}}{\partial \beta} &= 0 \\ \frac{\partial d_{1z}}{\partial \beta} &= -b_{1x} \cos \beta - b_{1y} \sin \alpha \sin \beta - b_{1z} \cos \alpha \sin \beta \end{aligned}$$

Thus:

$$\begin{aligned} \frac{\partial l_1}{\partial \alpha} &= \frac{1}{l_1} \left\{ d_{1x} (b_{1y} \cos \alpha \sin \beta - b_{1z} \sin \alpha \sin \beta) \right. \\ &\quad \left. + d_{1y} (b_{1y} \sin \alpha + b_{1z} \cos \alpha) \right. \\ &\quad \left. + d_{1z} (b_{1y} \cos \alpha \cos \beta - b_{1z} \sin \alpha \cos \beta) \right\} \\ \frac{\partial l_1}{\partial \beta} &= \frac{1}{l_1} \left\{ d_{1x} (-b_{1x} \sin \beta + b_{1y} \sin \alpha \cos \beta + b_{1z} \cos \alpha \cos \beta) \right. \\ &\quad \left. + d_{1z} (-b_{1x} \cos \beta - b_{1y} \sin \alpha \sin \beta - b_{1z} \cos \alpha \sin \beta) \right\} \end{aligned}$$

Analogously for d_2 :

$$\begin{aligned}
d_{2x} &= -a_{2x} + b_{2x} \cos \beta + b_{2y} \sin \alpha \sin \beta + b_{2z} \cos \alpha \sin \beta + p_x \\
d_{2y} &= a_{2y} - b_{2y} \cos \alpha + b_{2z} \sin \alpha - p_y \\
d_{2z} &= -a_{2z} - b_{2x} \sin \beta + b_{2y} \sin \alpha \cos \beta + b_{2z} \cos \alpha \cos \beta + p_z
\end{aligned}$$

$$\begin{aligned}
\frac{\partial l_2}{\partial \alpha} &= \frac{1}{l_2} \left\{ d_{2x} (b_{2y} \cos \alpha \sin \beta - b_{2z} \sin \alpha \sin \beta) \right. \\
&\quad + d_{2y} (b_{2y} \sin \alpha + b_{2z} \cos \alpha) \\
&\quad \left. + d_{2z} (b_{2y} \cos \alpha \cos \beta - b_{2z} \sin \alpha \cos \beta) \right\} \\
\frac{\partial l_2}{\partial \beta} &= \frac{1}{l_2} \left\{ d_{2x} (-b_{2x} \sin \beta + b_{2y} \sin \alpha \cos \beta + b_{2z} \cos \alpha \cos \beta) \right. \\
&\quad \left. + d_{2z} (-b_{2x} \cos \beta - b_{2y} \sin \alpha \sin \beta - b_{2z} \cos \alpha \sin \beta) \right\}
\end{aligned}$$

References

- [1] K. Lynch and F. Park, *Modern Robotics: Mechanics, Planning, and Control*. Cambridge University Press, 2017.